

I. Link Definition and Desired Margin

- a. **Next communication node:** the nearer of the two orbital relay agents. The rover's mission doesn't route through a direct link to Earth; Section VI works out why. This report follows the assignment's naming: rover $\rightarrow$ relay counts as uplink, since that's the direction the science data comes back on, and relay $\rightarrow$ rover counts as downlink, carrying commands out to the rover.
- b. **Band and channel:** UHF proximity link on the CCSDS Proximity-1 return/forward pair, 401.6 MHz return and 437.1 MHz forward, the same split flown by Electra and Electra-Lite on the Mars Relay Network.
- c. **Desired margin:** 3 dB above the E_b/N_0 required for a post-decoding bit error rate of 10^{-6} . CCSDS turbo coding at rate 1/2 puts that threshold near 1.5 dB, so the link has to clear 4.5 dB total. Three decibels is the right cushion given how much is still uncertain before flight. Nobody has flown the relay's exact orbit geometry yet. The gain pattern on the rover's helix is still just a model; nobody has range-tested the flight unit. And Jupiter's synchrotron emission adds an amount of receiver noise that is still a rough guess at this stage.
- d. **Worst-case geometry:** both relays fly a 200 km circular Europa orbit. With $R_{\text{Europa}} = 1561$ km and a 10° elevation mask, the slant range at the edge of the pass works out to

$$R = -R_E \sin \varepsilon + \sqrt{R_E^2 \sin^2 \varepsilon + (r^2 - R_E^2)} = 588 \text{ km},$$

compared with 200 km at zenith. Every budget from here on uses the 588 km worst case.

II. Assumed Parameters

- a. $P_t = 5$ W RF (rover), 8 W RF (relay). The rover's UHF radio draws 35 W DC in the power budget, and an Electra-Lite class transceiver puts about 5 W of that onto the antenna. The relay's power budget has more slack than the rover's, so it carries a larger amplifier.
- b. $G_t = 3.16$ (5.0 dBi) rover quadrifilar helix; $G_r = 5.01$ (7.0 dBi) relay UHF patch array. The helix already has a spot in the mass budget, and its near-hemispherical pattern means the rover can close a pass without pointing at anything.
- c. $L_a = 0.89$ (−0.51 dB). Europa's atmosphere, at roughly 10^{-12} bar, is far too thin to absorb anything here. This loss term comes from circular-to-linear polarization mismatch and Jovian plasma scintillation along the path.
- d. $L_l = 0.71$ (−1.49 dB), the transmitter-to-antenna line loss. It comes from the cabling and connectors running out to a mast-mounted antenna, derated for operation at a cryogenic 90 K.
- e. $\lambda = 0.7465$ m return (401.6 MHz), $\lambda = 0.6859$ m forward (437.1 MHz), from $\lambda = c/f$.
- f. $k = 1.38 \times 10^{-23}$ J/K.
- g. $T_s = 3000$ K at the relay receiver, 4000 K at the rover receiver. This is the number that really separates a Europa proximity link from a Mars one: at 400 MHz the receiver picks up galactic background noise on top of Jupiter's synchrotron emission, and the rover's front end, with no cryogenic cooling and a warmer view of the sky than the relay's, ends up carrying the higher figure.

- h. $R = 5.879 \times 10^5$ m, the 10° -elevation slant range derived above.
- i. $R_b = 128$ kbps return, 8 kbps forward. REQ.10.01 sets the floor here: getting 55 MB of compressed data down inside the 90 min relay window the ConOps allocates takes about 90 kbps, and 128 kbps is the next standard rate above that.

III. Method

- a. Ratio form:

$$\frac{E_b}{N_0} = \frac{P_t G_t G_r L_a L_l \lambda^2}{(4\pi R)^2 k T_s R_b}$$

- b. Decibel form: $(E_b/N_0)_{\text{dB}} = 10 \log_{10} (E_b/N_0)$
- c. Margin: $M = (E_b/N_0)_{\text{dB}} - (E_b/N_0)_{\text{req}}$, with $(E_b/N_0)_{\text{req}} = 1.5$ dB.
- d. The link closes when $M \geq 3$ dB.

IV. Uplink: Rover to Relay (Science Return)

- a. Numerator:

$$P_t G_t G_r L_a L_l \lambda^2 = 5 \times 3.16 \times 5.01 \times 0.89 \times 0.71 \times (0.7465)^2 = 27.87$$

- b. Denominator:

$$(4\pi R)^2 k T_s R_b = \left(4\pi \times 5.879 \times 10^5\right)^2 \times 1.38 \times 10^{-23} \times 3000 \times 1.28 \times 10^5 = 0.2892$$

- c. Result: $E_b/N_0 = 27.87/0.2892 = 96.4$
- d. In decibels: $10 \log_{10}(96.4) = 19.84$ dB
- e. Margin: $19.84 - 1.5 = 18.3$ dB. **The uplink closes.**

Here's the same result broken down term by term, in decibels, so each contribution is traceable:

Term	Value (dB)
P_t (5 W)	+6.99
G_t (rover helix)	+5.00
G_r (relay patch array)	+7.00
L_a (polarization + plasma)	-0.51
L_l (line loss)	-1.49
λ^2	-2.54
$(4\pi R)^2$ (space loss at 588 km)	-137.37
k	+228.60
T_s (3000 K)	-34.77
R_b (128 kbps)	-51.07
E_b/N_0	+19.84

V. Downlink: Relay to Rover (Command Forward)

a. Numerator:

$$8 \times 5.01 \times 3.16 \times 0.89 \times 0.71 \times (0.6859)^2 = 37.65$$

b. Denominator:

$$\left(4\pi \times 5.879 \times 10^5\right)^2 \times 1.38 \times 10^{-23} \times 4000 \times 8.0 \times 10^3 = 0.02410$$

c. Result: $E_b/N_0 = 37.65/0.02410 = 1562$

d. In decibels: $10 \log_{10}(1562) = 31.94 \text{ dB}$

e. Margin: $31.94 - 1.5 = 30.4 \text{ dB}$. **The downlink closes.**

The forward direction has it easier. The relay pushes more power out of a higher-gain antenna, and it only has to move 8 kbps of commands against 128 kbps of science data coming back the other way. That $16 \times$ difference in rate by itself is worth 12 dB of the gap.

VI. Direct-to-Earth Check (Why the Relay Exists)

Running the same equation on the rover's X-band low-gain antenna against a DSN 34 m BWG station gives the fallback rate. With $P_t = 15 \text{ W}$, $G_t = 6.31$ (8 dBi LGA), $G_r = 6.6 \times 10^6$ (68.2 dBi), $\lambda = 0.0357 \text{ m}$ (8.4 GHz), and $T_s = 30 \text{ K}$ (a cryogenically cooled DSN front end), solving for the rate that still holds 4.5 dB gives:

Earth range	R (m)	Max rate at 4.5 dB
Opposition (4.2 AU)	6.28×10^{11}	6.9 bps
Conjunction (6.2 AU)	9.27×10^{11}	3.2 bps

Seven bits per second covers a state-of-health beacon. One 55 MB sol of REQ.10.01 data would take just over two years to clear at that rate, which is why the relay agents belong in the design from the start. REQ.10.03 fixes this same number as the 5 bps fallback floor.

VII. Reflection

Both directions works, and by a wide margin: 18.3 dB on the return, 30.4 dB on the forward. The proximity link seems to run out of headroom somewhere other than the RF budget. At 4.5 dB the return could carry something like 4.4 Mbps, well past what the hardware and the pass geometry allow: the radio tops out at 2048 kbps, and the pass itself only lasts 90 min. That surplus is worth utilizing, since T_s is the weakest assumption in this analysis, and Jupiter's synchrotron noise at 400 MHz could run several times higher than the 3000 K figure I used. If the link failed anyway, R_b is the first thing I'd change, since adjusting it costs nothing. G_r at the relay comes next. Adding patch elements to an orbiter costs less in mass and power than any equivalent change on the rover.

References:

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- [2] Consultative Committee for Space Data Systems, “Proximity-1 Space Link Protocol — Physical Layer,” CCSDS 211.1-B-4, 2013.
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- [4] C. D. Edwards *et al.*, “The Electra Proximity Link Payload for Mars Relay Telecommunications and Navigation,” Jet Propulsion Laboratory, Pasadena, CA.
- [5] NASA, “Deep Space Network,” NASA Space Communications and Navigation. [Online]. Available: <https://www.nasa.gov/communicating-with-missions/dsn/>.
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AI disclosure: I used AI to check the arithmetic behind each link budget term and to cross-check the assumed parameters against published Electra and DSN performance data. I picked the architecture and the parameters myself, and I set the margin the same way.